

LOCAL ANESTHETICS

PROF.DR. RIFFAT FAROOQUI
HEAD DEPT. OF PHARMACOLOGY
DIKIOHS

LA

**BLOCK NERVE CONDUCTION OF SENSORY IMPULSES
AND IN HIGHER CONCENTRATIONS BLOCK MOTOR
IMPULSES FROM THE PERIPHERY TO THE CNS BY
PREVENTING PROPAGATION OF ACTION POTENTIAL**

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CLASSIFICATION

ESTERS:

COCAINE

PROCAINE

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CHLORPROCAINE

BENZOCAINE

TETRACAINE

AMIDES:

☐ LIGNOCAINE

☐ ROPIVACAINE

☐ MEPIVACAINE

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☐ BUPIVACAINE

☐ LEVOBUPIVACAINE

☐ PRILOCAINE

SHORT ACTING:

Procaine

Chlorprocaine

INTERMEDIATE ACTING:

Lignocaine

Prilocaine

LONG ACTING:

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Bupivacaine

Levobupivacaine

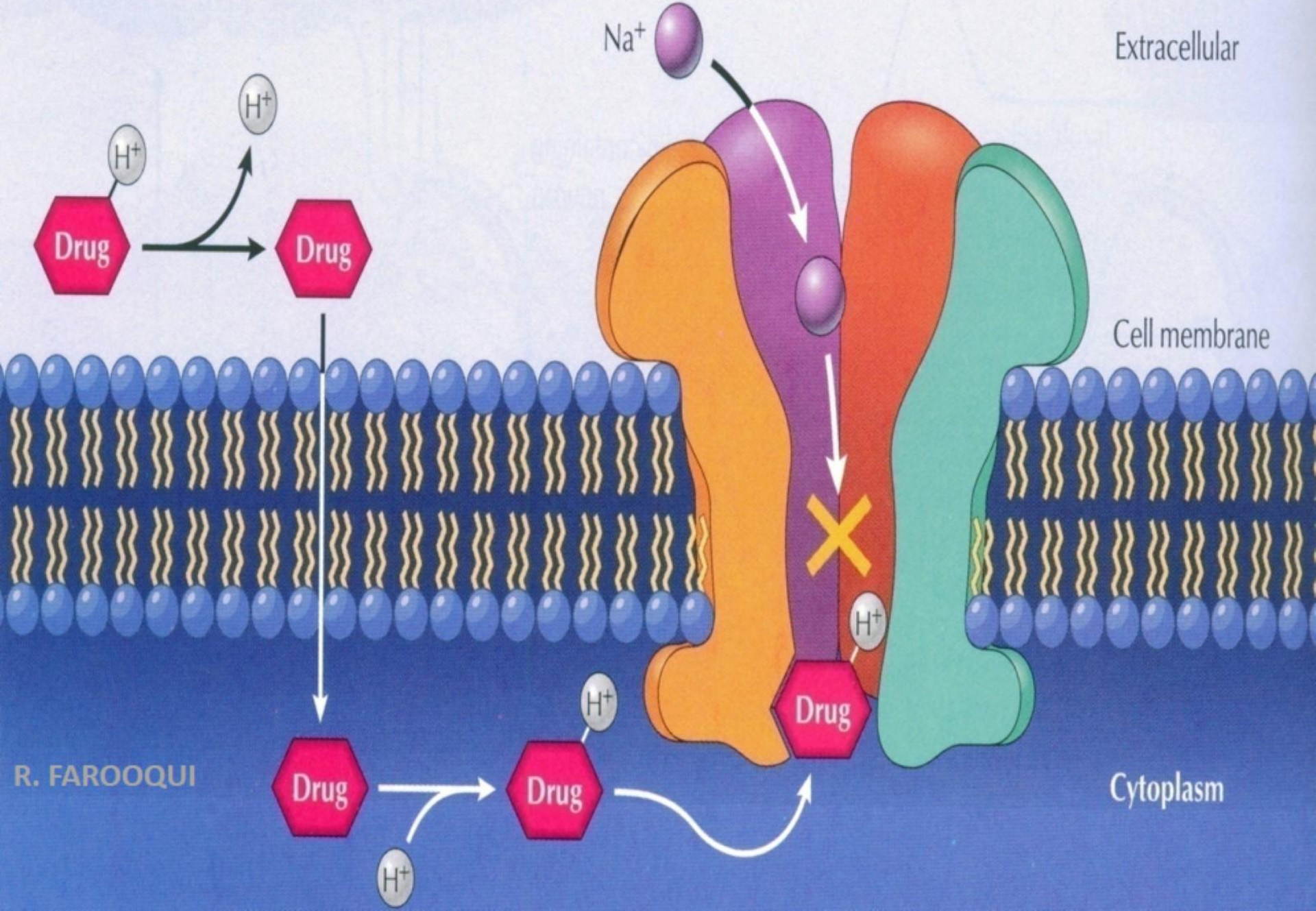
Ropivacaine

Tetracaine

MOA

- **TYPE OF FIBER: TYPE C AND TYPE B**
- **STATE OF CHANNEL: ACTIVATED** R. FAROOQUI
- **BLOCKAGE OF VOLTAGE GATED SODIUM CHANNEL IN RAPIDLY FIRING AXONES**
- **UNIONISED FORM PENETRATES NERVE CELL MEMBRANE IONIZED FORM INTERACTS WITH PROTEIN RECEPTOR OF SODIUM CHANNEL**
- **SLOW DOWN NERVE IMPULSE CONDUCTION**
- **ABOLISH GENERATION OF ACTION POTENTIAL**

Local Anesthetic Mechanism of Action

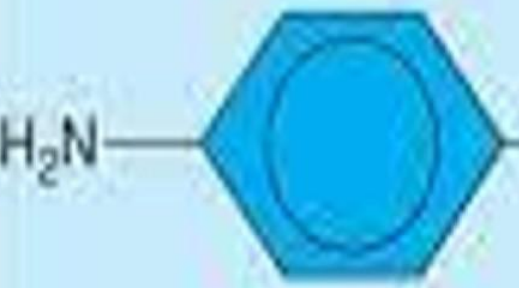
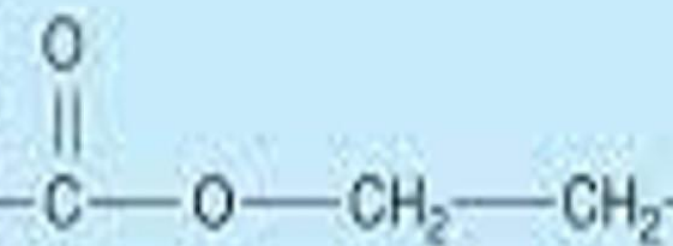
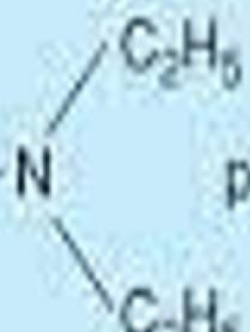
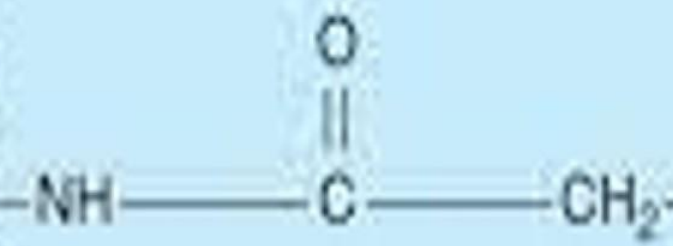
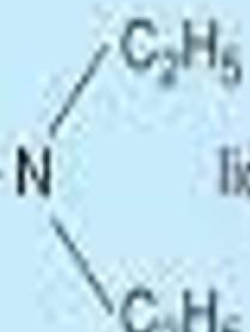


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Important point

- The **unionized form** of the local anesthetic can penetrate the nerve membrane.
- After entering the nerve, the drug exists in an ionized form that interacts with the Na^+ channel.
- The drug therefore needs to reach the nerve membrane and then interact with the channel to produce blockade.
- **Small fibers are blocked more readily than large fibers.**
- Therefore, sensory blockade generally occurs before significant motor blockade.

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aromatic part	intermediate chain	amino group
lipophilic 		hydrophilic  procaine
		 lidocaine

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Actions of local anesthetics

Vasodilation

causes:

- **↑ blood flow → ↑ diffusion/absorption away from injection site → shorter duration**
- Therefore, when a local anesthetic is administered alone, its effect may be relatively short.

Addition of epinephrine

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- **Local anesthetic + epinephrine**

Epinephrine causes vasoconstriction at the site.

Therefore:

- **↓ absorption/diffusion → ↑ duration of local anesthesia**
- It can also reduce systemic toxicity by slowing systemic absorption.

Onset of action

Factors influence the onset of local anesthetic action:

- Tissue pH
- Nerve morphology
- Drug concentration
- **pKa**
- Lipid solubility

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Most important factor: pKa

- A local anesthetic with a **lower pKa** has a greater proportion of drug in the unionized form at physiologic pH.
- The unionized drug can penetrate the nerve membrane more readily.

Therefore:

- **Lower pKa → more unionized drug → faster membrane penetration → faster onset**

Effect of tissue pH

- In infected/inflamed tissue:
- **↓ tissue pH → more acidic environment**
- This changes the ionization of the local anesthetic.
- Less drug is available in the form that can readily penetrate the nerve membrane.
- Therefore:
- **Acidic/infected tissue → delayed or sometimes prevented onset of local anesthesia.**

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Remember

- **Infection → acidic tissue → local anesthetic works poorly.**

ONSET OF ACTION

PKa value LA: 7.7-9.1 (More Unionized Form Exists Allow Good Penetration Of Nerve Cell Membrane)

LOW Pka Value: Rapid onset of action

e.g: BENZOCAINE (Pka : 2.8): 1min

LIDOCAINE (Pka: 7.8): 2-4 min

BUPIVACAINE (PKa: 8.1): 5- 8 min

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LIPOHILICITY:

MORE LIPOPHILIC HIGH POTENCY

BUPIVACAINE HIGHLY LIPOPHILIC

POTENCY: 0.1 % TO 0.5%

LIDOCAINE LESS LIPOPHILIC

POTENCY: 1% TO 2%

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DURATION OF ACTION

LENGTH OF ESTER OR AMIDE CHAIN/DEGREE OF PROTEIN BINDING

BUPIVACAINE LONGER AMIDECHAIN LONGER
DURATION OF ACTION (4 HOURS)

PROTEIN BINDING: 95%

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- REASON OF SHORT DURATION OF ACTION:
VASODILATION
- ADDITION OF WHICH DRUG INCREASES
DURATION OF ACTION: EPINEPHRINE
- DURATION OF ACTION DEPENDENT ON:
REDISTRIBUTION NOT METABOLISM

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PHARMACOKINETICS

- ❑ METABOLISM OF ESTERS (PLASMA PSEUDOCHOLINESTRASE)
- ❑ METABOLISM OF AMIDES (LIVER/ P-450 HYDROLYSIS)
- ❑ ROLE OF EPINEPHRINE (DECREASE ABSORPTION INTO SYSTEMIC CIRCULATION, INCREASES DURATION OF ACTION, DECREASES SYSTEMIC TOXICITY)
- ❑ PRILOCAINE METABOLISED IN PLASMA AND KIDNEY ITS METABOLITE CAN CAUSE: METHMOGLOBINEMIA
- ❑ ESTERS METABOLITE CAUSE ALLERGIC REACTION : PABA
- ❑ ALLERGY TO AMIDE IS RAE

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ADRs: CNS

- **TONGUE NUMBNESS**
- **NYSTAGMUS**
- **MUSCULAR TWITCHING**
- **CONVULSIONS**
- **HALLUCINATIONS & ADDICTION (COCAINE)**
- **NEUTOPARHIC SYMPTOMS**

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CARDIOVASCULAR SYSTEM

- VASODILATATION
- HYPOTENSION
- CARDIAC ARRHYTHMIAS (COCAINE)
- PRILOCAINE LARGE DOSES
- METHEMOGLOBINEMIA
- CV COLLAPSE

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**❑ COCAINE: INTRINSIC SYMPATHOMIMETIC
ACTIVITY**

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**❑ BUPIVACAINE: CARDIAC TOXICITY (HIGH
DOSE)**

❑ LIDOCAINE: MORE NEUROTOXIC

LAST

- ☐ ALTERED MENTAL STATUS
- ☐ SEIZURES
- ☐ CV INSTABILITY

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TREATMENT OF LAST:

- ☐ SEIZURE SUPPRESSION
- ☐ CP SUPPORT
- ☐ LIPID RESCUE THERAPY

CLINICAL USES

- ☐ DENTAL PROCEDURES
- ☐ TOPICAL ANESTHESIA
- ☐ NEUROPATHIC PAIN
- ☐ ACUTE PAIN IN THE PERIOPERATIVE
PROCEDURE
- ☐ CHRONIC PAIN
- ☐ ARRHYTHMIAS (IV LIDOCAINE)

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TECHNIQUES OF LA

- ❑ **TOPICAL ANESTHESIA**

- ❑ **INFILTRATIVE ANESTHESIA**

- ❑ **PERINEURAL AND NEURAXIAL BLOCK**

- **CAUDAL**

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- **EPIDURAL (EXTRADURAL)**

- **SPINAL**

Technique	Basic concept
Topical	Applied to surface/mucosa
Infiltration	Injected into tissues
Peripheral nerve block	Injected near a nerve
Clinical techniques — high-yield table	
Spinal	Injection into neuraxial/subarachnoid region
Epidural	Injection into epidural space
Caudal	Neuraxial administration through caudal approach

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Neuraxial anesthesia

Includes:

- **Spinal anesthesia**
- **Epidural anesthesia**
- **Caudal anesthesia**
- These techniques produce regional anesthesia by acting on nerves associated with the spinal/neuraxial region.

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INTOLERANCE

- **ALLERGIC REACTIONS /
ANAPHYLACTIC REACTIONS
ESTERS (PABA METABOLITE)**

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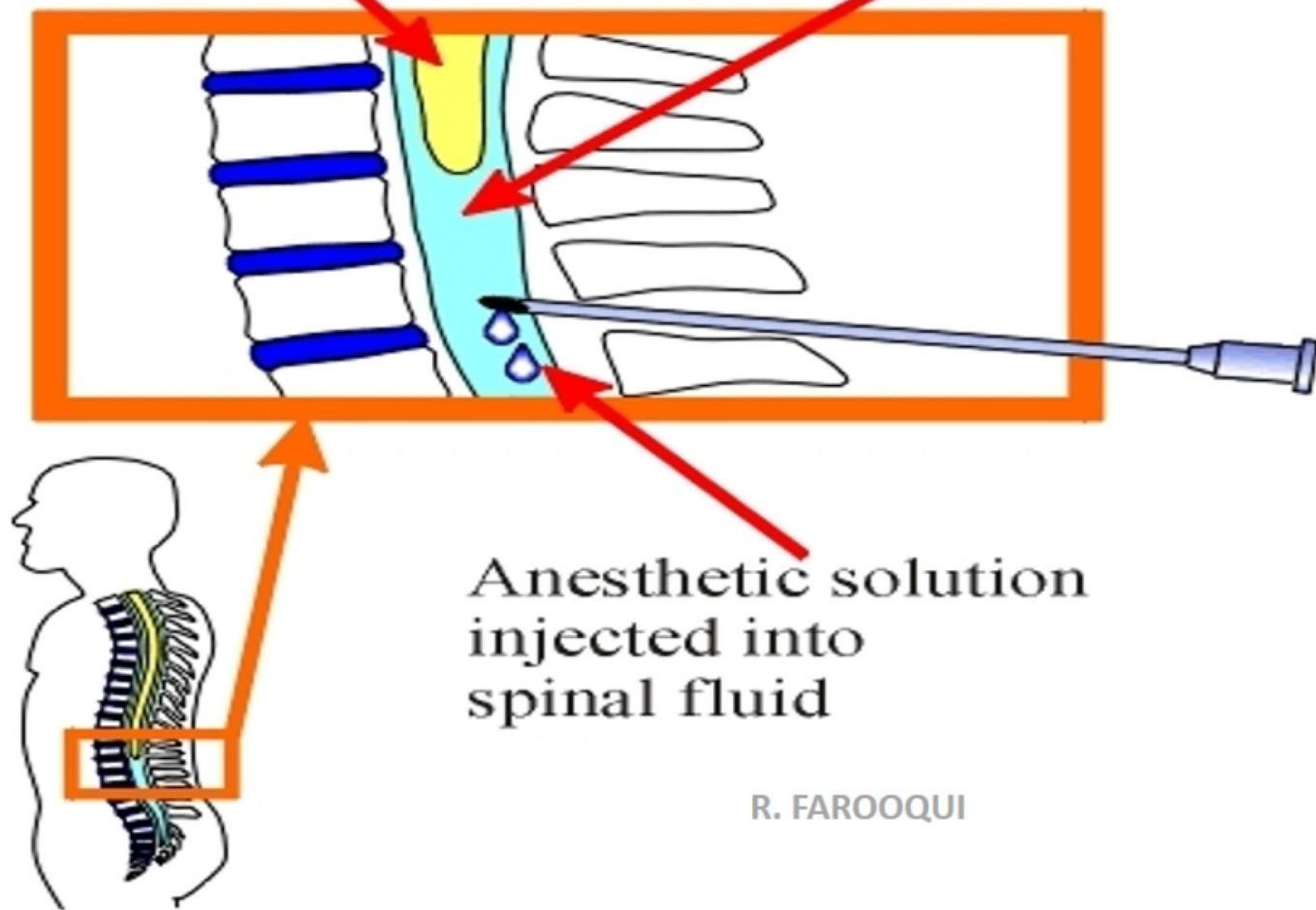


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Spinal Cord

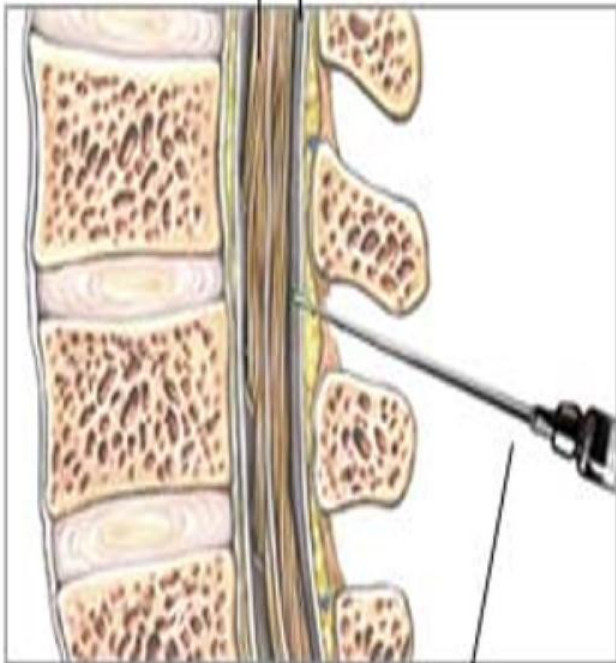
Spinal Fluid



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Spinal column

Epidural space



Regional anesthesia
(epidural)

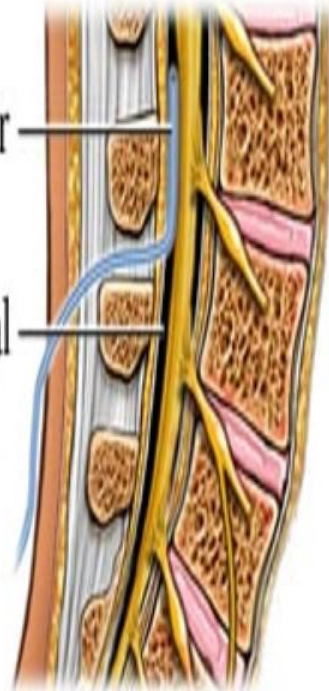


Desensitized
region



Catheter

Epidural
space



BRAIN WARM-UP QUIZ1.1

1. Which **mechanism** best describes the action of local anesthetics?
2. Which has **high surface local anesthetic** activity and **intrinsic vasoconstrictor** action?
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3. Which **factor** is the **most important determinant of the onset** of action of a local anesthetic?
4. Which drug is classified as an **amide local anesthetic**?

5. A patient requires an anesthetic for labor epidural analgesia. Which drug is specifically highlighted this purpose?
6. Which naturally occurring compound is a local anesthetic?
7. The primary metabolism of ester local anesthetics occurs through?
8. The clinical duration of local anesthesia is primarily related to?
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9. Adding epinephrine to a local anesthetic generally?
10. Which local anesthetic is associated with extended-duration anesthesia?
11. Which nerve fibers are particularly sensitive to local anesthetics?

OPTIONS

- A. Blockade of voltage-gated Ca^{2+} channels
- B. Blockade of voltage-gated Na^{+} channels
- C. Protein binding
- D. Lipid solubility
- E. pKa
- F. Procaine
- G. Lidocaine
- H. Bupivacaine
- I. Cocaine
- J. Plasma cholinesterase
- K. Monoamine oxidase
- L. Hepatic metabolism at the injection site
- M. Redistribution away from the site
- N. Decreases absorption and prolongs its action
- O. Increases its systemic absorption
- P. Large motor fibers only
- Q. Small fibers carrying pain and temperature

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THANK YOU